GC Kiosk Dispensing Routine: Explanation

Overview:

The primary goal of the kiosk is to dispense the correct number of sheets, ranging from 1 to 10, as configured through the portal. This process relies heavily on optical sensors to track and count each sheet as it is dispensed. The software must ensure accurate counting of the sheets using these sensors and handle various conditions that may arise during operation.

Components and Sensors:

1. **Optical Sensors**:
    - Two optical sensors are used on each dispense chute: one for scented sheets (left side) and one for unscented sheets (right side). Each chute has both a **top** and **bottom** sensor.
    - These sensors count sheets as they pass through. Each sheet breaks the IR beam of the optical sensor as it drops into the chute. When the beam is broken, the system registers the sheet, and the motor continues until the desired number of sheets has been dispensed.

Dispensing Logic:

1. **Sheet Counting Logic**:
    - The system must count the sheets using the optical sensors. Each time a sheet breaks the beam of the sensor, the counter increments. Once the count equals the configured sheet dispense number, the motor stops.
    - The **top sensor** detects when the sheet passes into the chute, and the **bottom sensor** confirms that the sheet has fully dropped.
2. **Motor Timeout Logic**:
    - Motor timeout acts as a safety mechanism, which should activate only in exceptional cases (10-20% of the time).
    - The timeout needs to be dynamic. For instance, if the system is set to dispense 5 sheets, the system should know it takes around 5 to 8 seconds to dispense them.
    - The software should adjust the motor timeout based on the number of sheets being dispensed and use sensor feedback to determine whether to extend the timeout if the process is taking longer than expected.

Action Steps for Dispensing:

1. **Initial Check**:
    - Before starting the dispensing process, check the status of the optical sensors. Ensure none are currently triggered (i.e., nothing is blocking the chute).
    - If a sensor is triggered unexpectedly, a separate routine must handle this condition (e.g., a sheet might be stuck or a partial sheet is still in the system).
2. **Dispense Routine**:
    - If all sensors are clear:
        - Start the motor and begin dispensing sheets.
        - Use the top and bottom sensors to count each sheet as it drops.
    - If one or more sensors are triggered before dispensing:
        - Check for possible issues such as a misaligned sensor or a sheet stuck in the chute.
        - Adjust the routine to resolve the issue before continuing.

Potential Issues with Optical Sensors:

- **Partial sheets**: If a sheet is not fully dispensed, it may block the top sensor.
- **Misaligned sensors**: If the sensor is misaligned, it won't correctly detect the sheets, requiring physical adjustment.
- **Obstructions in the chute**: A sheet may become stuck in the chute, blocking the sensor.

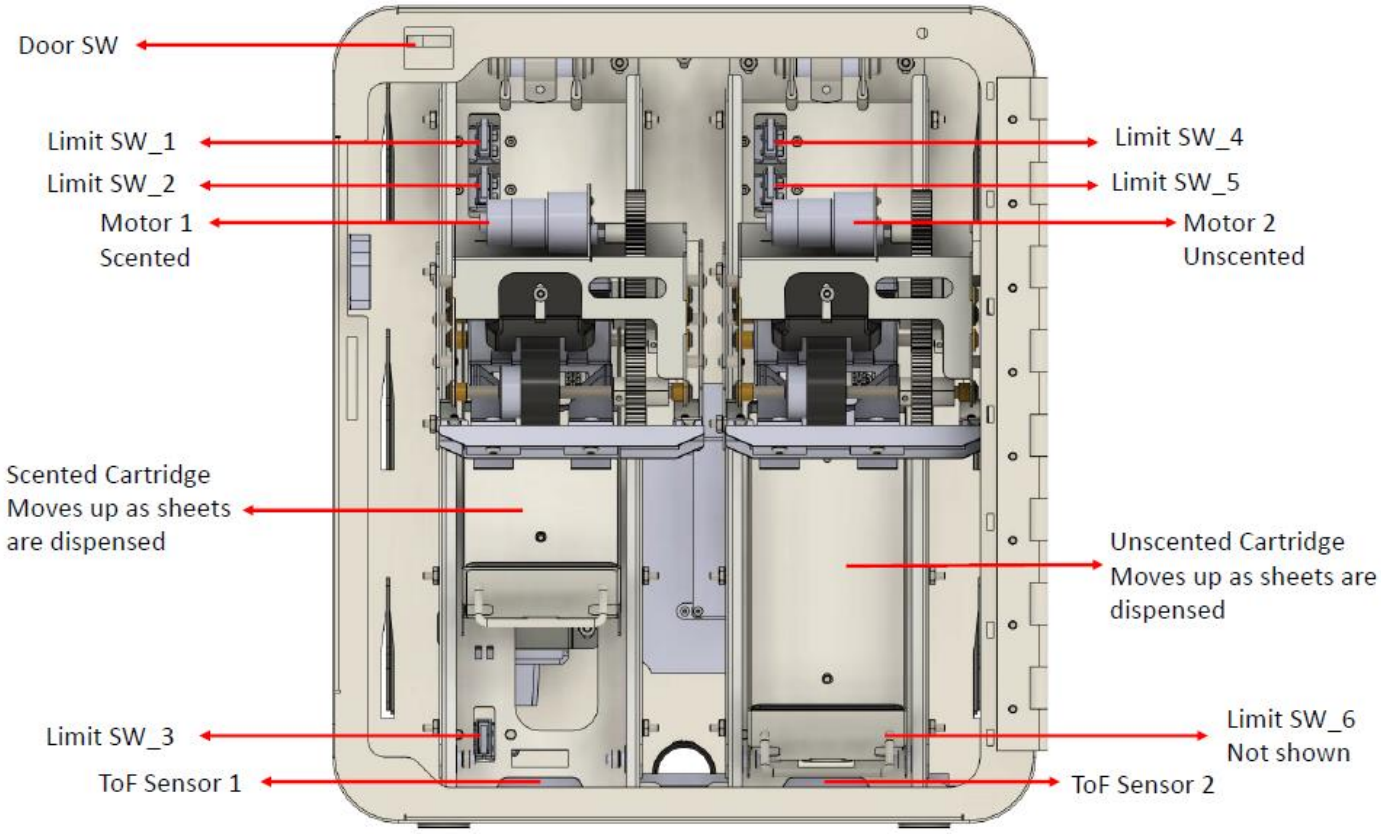
The motor timeout logic should consider the expected time for dispensing and adjust according to real-time sensor feedback to ensure accurate operation.

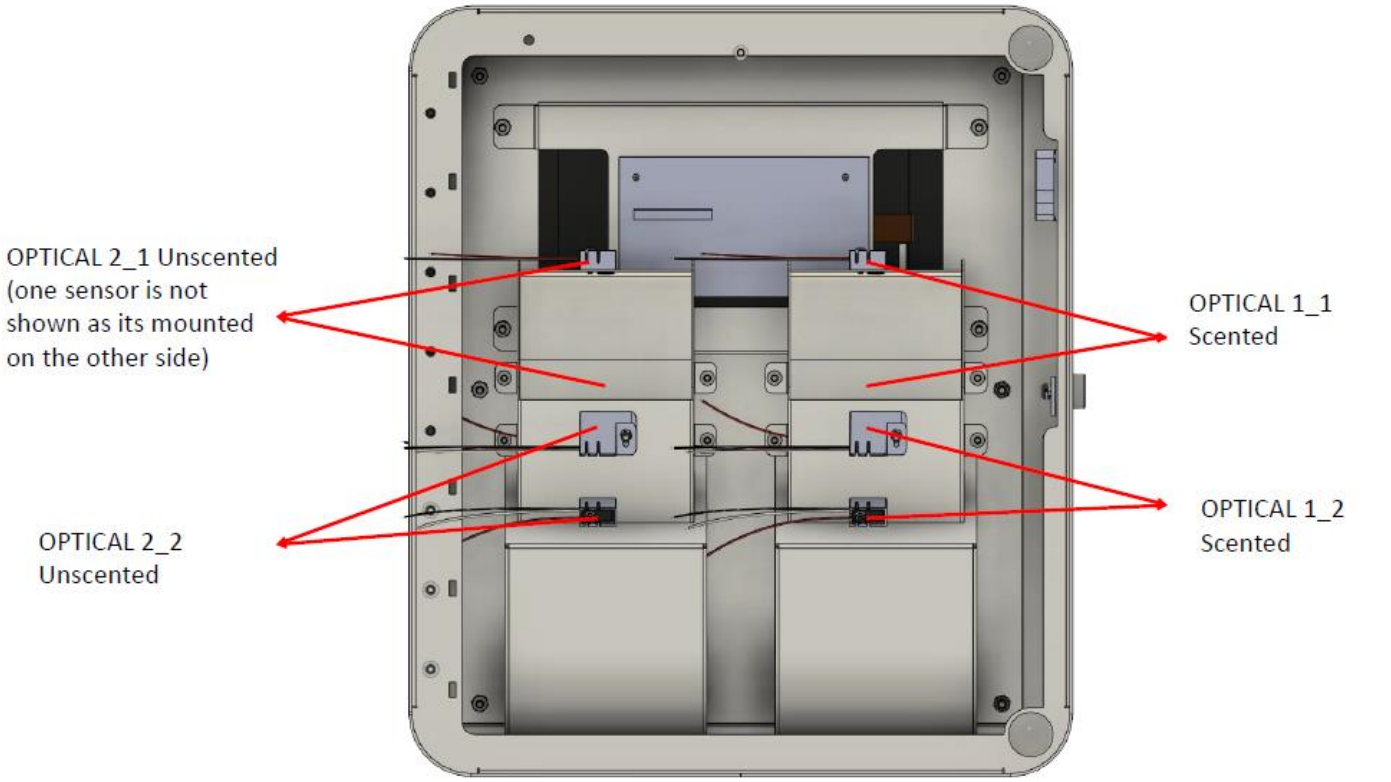
Distance Measurement and Inventory Management:

1. **Time of Flight (ToF) Sensors**:
    - ToF sensors are used to estimate the inventory of sheets in the tray. As sheets are dispensed, the tray moves up, increasing the distance between the ToF sensor and the sheets.
    - Inaccurate ToF data can occur, with error codes like 291, indicating the data is unavailable. When this happens, the system should bypass the ToF check and rely solely on optical sensors to determine if sheets are available for dispensing.

Handling Errors:

1. **When ToF Data is Unavailable**:
    - If ToF data returns an error, the system must still attempt to dispense using only optical sensor data. The system should count sheets and adjust based on optical sensor feedback, bypassing ToF dependency.
2. **Timer-Based Backup**:
    - If sensor data is unavailable, a timer-based backup routine can be implemented to ensure that the motor runs for a reasonable amount of time, checking periodically for sheet dispensing completion.


Door SW
Limit SW_1
Limit SW_2
Motor 1
Scented
Limit SW_4
Limit SW_5
Motor 2
Unscented
Scented Cartridge
Moves up as sheets
are dispensed
Unscented Cartridge
Moves up as sheets are
dispensed
Limit SW_3
ToF Sensor 1
Limit SW_6
Not shown
ToF Sensor 2


OPTICAL 2_1 Unscented
(one sensor is not
shown as its mounted
on the other side)
OPTICAL 1_1
Scented
OPTICAL 2_2
Unscented
OPTICAL 1_2
Scented